🎉 Faculty Achievement Spotlight 🎉

🌟 Congratulations to Our Esteemed Faculty! 🌟

📸 Faculty Photos:



Name of the Faculty: Dr. R. VENKATESAN,

Designation: Associate Professor

Department: Department of Biomedical Engineering

Achievement: Research Article Publication

📌 Publication/Work Details

📖 **Title**: Design, Modeling, and Parametric Optimization of Carbon Nanotube Thin-Film Transistors for High-Performance Display Applications

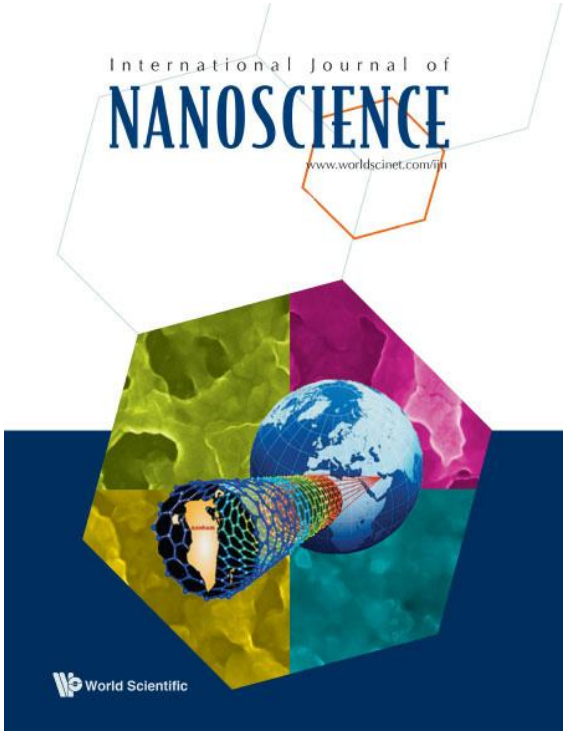
Name of the journal/ Publisher: International Journal of Nanoscience/ World Scientific Publishing Company

Month and year of publication: 29-10-2026

Indexed In: ESCI ,Scopus, and SCImago

Type of Work: Research Article

Snapshot of the Work:

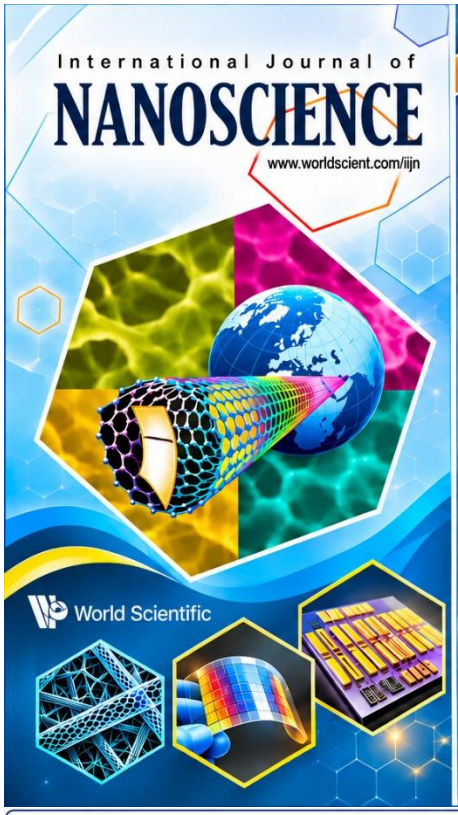




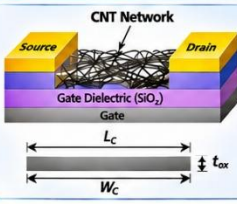
Design, Modeling, and Parametric Optimization of Carbon Nanotube Thin-Film Transistors for High-Performance Display Applications

Abstract

Carbon nanotubes (CNTs) have attracted significant attention as semiconductor channel materials for thin-film transistor (TFT) applications due to their one-dimensional geometry, superior carrier mobility, and favorable electrical properties. While TFTs are integral components of contemporary display technologies, conventional TFT materials continue to suffer from drawbacks such as high processing temperatures, limited charge transport capability, low ON/OFF current ratios, and poor thermal stability. In this study, CNT-based thin-film transistors (CNTTFTs) are systematically designed, simulated, and optimized using the Nanonet simulation platform, yielding device characteristics that outperform those of traditional TFT configurations. The simulation results demonstrate substantial variation in drain-current performance with CNT length, channel length, network density, and nanotube orientation, with higher current obtained under favorable CNT-network configurations. The results establish the influence of the CNTlength- to-channel-length ratio and network parameters on conductive-path formation and provide design guidance for CNT-TFT optimization. Furthermore, the influence of key CNT physical parameters on transistor current behavior is examined in detail. The results demonstrate that shortchannel operation facilitates enhanced current scaling, and that device performance is predominantly controlled by the ratio of CNT length to inter-tube spacing rather than by the absolute nanotube length. It is also observed that CNT orientation angles of 30° and 60° provide improved transport stability and uniform electrical performance. In addition to improved electrical characteristics, CNT-TFTs show strong potential as driver elements in next-generation flat-panel display (FPD) systems by delivering high drive currents while maintaining thermally safe operation. These findings establish practical design considerations, identify existing limitations, and provide valuable insight for future optimization and fabrication of CNT-based TFT devices. Overall, the study strengthens the understanding of structure–property relationships in CNT-TFTs and confirms their promise as high-mobility semiconductor channels for advanced electronic applications.



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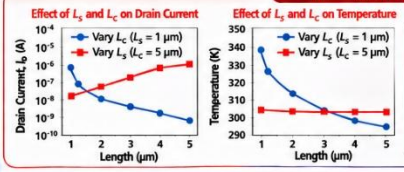


Key Parameters

- L_S : Tube length
- L_C : Channel length
- W_C : Channel width
- D : Network density
- θ : Tube orientation

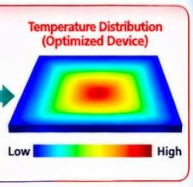


KEY RESULTS



Optimal Design Window

- $L_S \approx L_C$ (4 – 5 μm)
- $\theta \approx 60°$
- D = 12 – 15 tubes/μm²
- High I_D with Near Room-Temperature Operation



CNT-TFT Applications: Flat Panel Displays | Flexible Electronics | Wearable Sensors | Low-Power Nanoelectronics

International Journal of Nanoscience
ISSN (Print): 0219-581X
ISSN (Online): 1793-5350
DOI: 10.1142/S0219581X26400508

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Thermal-Aware Optimization of CNT-TFT Device Parameters for High Current and Room-Temperature Operation in Flat Panel Displays

MOTIVATION

CNT-TFTs are promising for next-generation flat panel displays due to high mobility and flexibility.



Design Challenge
Achieve high drain current while maintaining safe operating temperature (~Room Temp.)

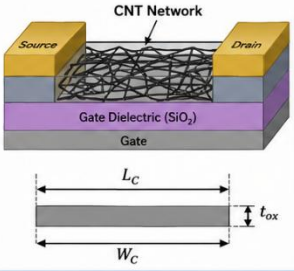
OUR APPROACH


Percolation-based CNT network simulation


Electro-thermal coupled analysis


Optimize device parameters for high current with low temperature

CNT-TFT DEVICE STRUCTURE



Key Parameters

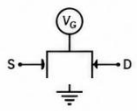
- L_S : Tube length
- L_C : Channel length
- W_C : Channel width
- D : Network density
- θ : Tube orientation

OPTIMIZATION METHODOLOGY

1. Vary Key Parameters

- L_S = 1 – 5 μm
- L_C = 1 – 5 μm
- D = 10 – 30 tubes/μm²
- θ = 30°, 60°, 90°

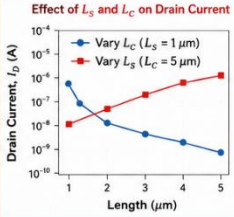
2. Evaluate Performance

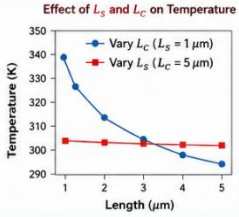


- Drain Current (I_D)
- Temperature (T)

3. Identify Optimal Combination for High I_D and Near Room-Temperature Operation

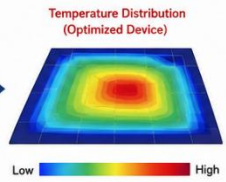
KEY RESULTS





Optimal Design Window

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CONCLUSION

Thermal-aware optimization reveals that CNT-TFTs with $L_S \approx L_C$ (4 – 5 μm), $\theta \approx 60°$ and moderate network density achieve high drain current while maintaining near room-temperature operation. Simulation results agree well with reported experimental data.

→ High Drain Current → Near Room-Temperature Operation → Reliable & Stable Performance → Suitable for Flat Panel Display Backplanes